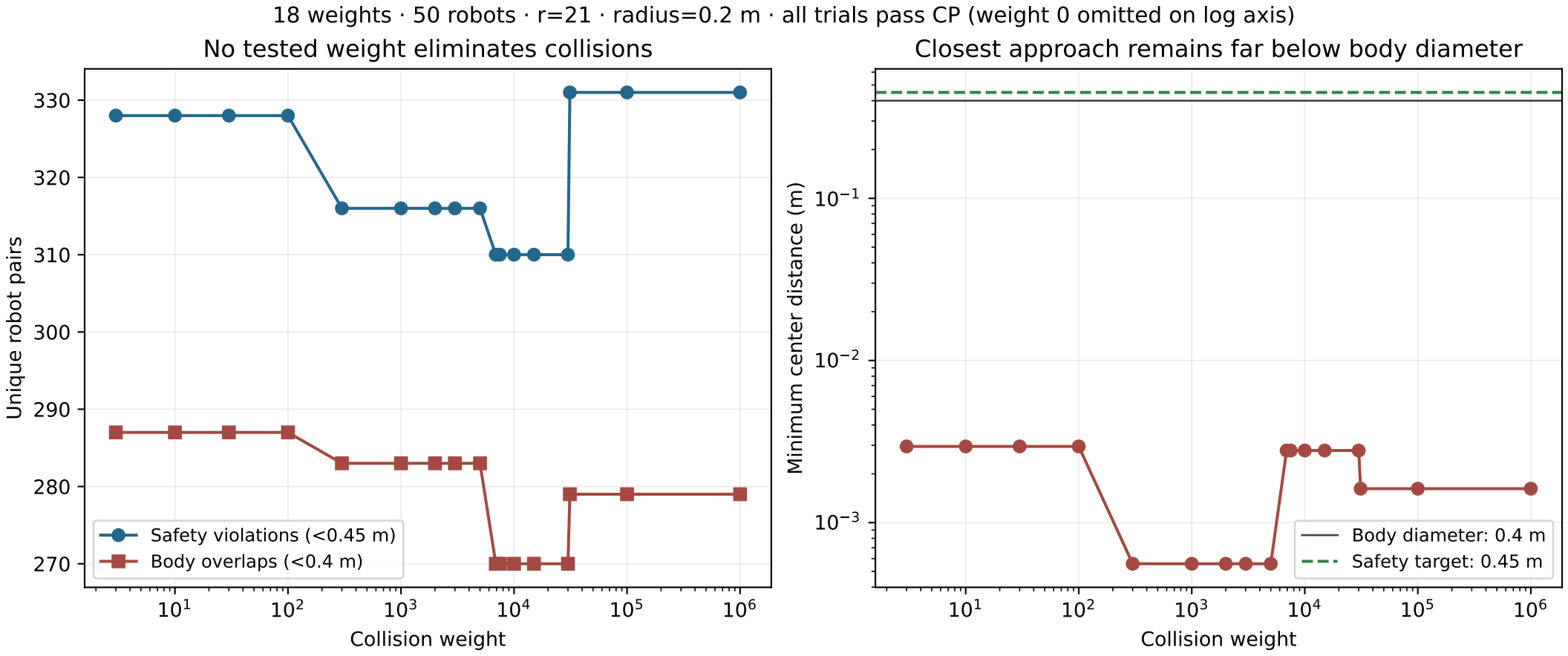
18 weights · 50 robots · r=21 · radius=0.2 m · all trials pass CP (weight 0 omitted on log axis)
No tested weight eliminates collisions
Unique robot pairs
330
320
310
300
290
280
270
Safety violations (<0.45 m)
Body overlaps (<0.4 m)
10^1
10^2
10^3
10^4
10^5
10^6
Collision weight
Closest approach remains far below body diameter
Minimum center distance (m)
10^{-1}
10^{-2}
10^{-3}
Body diameter: 0.4 m
Safety target: 0.45 m
Collision weight